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dlBNST CRF Neurons: Location, Markers, Subtypes, Connectivity, and Transcriptomic Diversity

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Background

The BNST is a limbic hub that links autonomic, endocrine, and behavioral control, and the dorsolateral BNST sits within a structure that is deeply heterogeneous. For dlBNST CRF neurons, this means their properties are best understood as part of a diverse and possibly continuous cell landscape rather than as a single uniform cell class. (5 sources)

The dorsolateral bed nucleus of the stria terminalis (dlBNST) should be viewed against the broader BNST, which is a heterogeneous forebrain structure that helps relay information between autonomic, neuroendocrine, and behavioral systems. The BNST contains more than 16 subregions and distinct neuronal populations defined by differences in transmitters, receptors, and neuropeptides, making any CRF-positive population inherently part of a complex local mosaic rather than a simple unitary class (Miura et al., 2023). More generally, work across neuroscience shows that even neurons grouped into the same broad type can differ strongly in molecular state and physiology, and that important variation also exists within a given neuronal type (Reva et al., 2025) (Eberwine et al., 2015). This point matters for dlBNST CRF neurons because functional classifications in BNST already suggest multiple neuronal types, yet these types may not form rigid bins and instead may fall along a continuum that would be better resolved by pairing physiology with transcriptomics (Miura et al., 2023). In parallel, single-cell transcriptomic work has emphasized that cell-to-cell RNA variation is not just noise but can carry functionally relevant information, and models built from single-cell expression can predict single-neuron physiology better than cell-type averages (Reva et al., 2025) (Kendrick et al., 2024). A useful conceptual backdrop is that neurons can preserve similar phenotypes despite variable underlying molecular balances, so stable functional output does not imply molecular uniformity (Reva et al., 2025) (Swensen et al., 2005). Together, this sets up the key idea for dlBNST CRF neurons: they are part of a stress-linked neuropeptidergic population embedded in a highly subdivided, heterogeneous BNST, and understanding their structure and function requires attention to both circuit position and within-class diversity (Miura et al., 2023) (Reva et al., 2025).

Evidence

(Miura et al., 2023)

Electrophysiologically distinct bed nucleus of the stria terminalis projections to the ventral tegmental area in mice

"The bed nucleus of the stria terminalis (BNST) is a highly heterogeneous limbic forebrain structure that serves as a relay

connecting autonomic, neuroendocrine and behavioral function. It can be divided into over 16 individual subregions with distinct neuronal subpopulations based on receptors, transmitters, and neuropeptides"

".While we have functionally identified four distinct neuronal types, a transcriptome analysis of the individual subtypes would help in our understanding of the rigidity of neuronal classification. This applies particularly when the functional type of BNST neurons might be seen as a continuum on a heatmap. Given a study showing that chronic social defeat stress downregulated the expression level of several ion channels in resilient mice (Gururajan et al., 2022), a heatmap classifying BNST neurons combined with transcriptomics will assist in visualizing how the population shifts."

(Reva et al., 2025)

Higher-order interactions in neuronal function: From genes to ionic currents in biophysical models

"While neurons are often categorized under broad classes, a number of studies have demonstrated that, even in seemingly homogeneous neuronal populations, neurons exhibit profound heterogeneity across multiple dimensions, ranging from genetic and molecular differences to variations in electrophysiological properties (1)(Eberwine et al., 2015)(Kendrick et al., 2024)(Swensen et al., 2005)(Schulz et al., 2006). Thus neuronal diversity appears as an intricate and multifaceted phenomenon, involving not only variations between neuronal types but also within neuronal types"

".In fact, single-cell transcriptomics has consistently uncovered significant variations in mRNA and noncoding RNA levels between individual neurons of the same neuronal type, emphasizing the significance of molecular heterogeneity."

(Eberwine et al., 2015)

Cellular Deconstruction: Finding Meaning in Individual Cell Variation

"The advent of single cell transcriptome analysis has permitted the discovery of cell-to-cell variation in transcriptome expression of even presumptively identical cells. We hypothesize that this variability reflects a many-to-one relationship between transcriptome states and a cell's phenotype. In this relationship the molecular ratios of the subsets of RNA are determined by the cell systems' stoichiometric constraints, which underdetermine the transcriptome state. Further the variability is in part induced by the tissue context and is important for system-level function. This theory is analogous to theories of literary deconstruction where multiple "signifiers" work in opposition to one another to create meaning. By analogy, transcriptome phenotypes should be defined as subsets of RNAs comprising selected RNA systems where the system associated RNAs are balanced with each other to produce the associated cellular function. This idea provides a framework for understanding cellular heterogeneity in phenotypic responses to variant conditions such as disease challenge."

(Kendrick et al., 2024)

Transcriptomically-measured gene expression predicts physiological variation across single neurons in humans and mice

"Single-cell transcriptomics measures the molecular landscape of individual neurons with unprecedented efficiency and scale. This insight has the potential to advance our understanding of the molecular basis of neuronal function, and to identify druggable targets for disease treatments. However, transcriptomics also suffers from greater measurement noise than traditional techniques (e.g., RT-PCR), which raises questions about its ability to offer insight into function at true single-cell resolution. We tested if transcriptomic data could yield insight into function of individual neurons in human and mouse neocortex by analyzing two datasets collected via Patch-Seq, a powerful technique for obtaining transcriptomic and physiology data from the same neuron. We found that computational models trained on single-cell transcriptomic data robustly predicted physiology of individual neurons. Critically, models trained on single cells outperformed those trained on cell type averages when predicting single-cell physiology. Thus, the standard approach of denoising single-cell transcriptomic data by averaging on cell types sacrifices functionally-relevant information. Our analysis also revealed novel relationships between gene expression and physiology, including a potential molecular substrate of human- mouse cross-species differences in the speed of single-neuron computation. Broadly, our findings highlight the promise of Patch-Seq for generating new insight into the molecular basis of neuronal function."

(Swensen et al., 2005)

Robustness of Burst Firing in Dissociated Purkinje Neurons with Acute or Long-Term Reductions in Sodium Conductance

"Cerebellar Purkinje neurons often generate all-or-none burst firing in response to depolarizing stimuli. Voltage-clamp experiments using action potential waveforms show that burst firing depends on small net inward currents that flow after spikes and reflect the net balance between multiple large currents. Given this, burst firing is surprisingly robust in the face of changes in the magnitude of the underlying currents from cell to cell. We explored the basis of this robustness by examining the effects of reducing the sodium current, the major contributor to the postspike inward current. Burst firing persisted in concentrations of tetrodotoxin that produced half-block of sodium current. This robustness of bursting reflects an acute feedback mechanism whereby waveform changes from the reduced sodium current (reduced spike height and a hyperpolarizing shift in postspike voltage) cause compensatory decreases in postspike potassium currents. In particular, reduced spike height reduces calcium entry and subsequent calcium-activated potassium current, and the hyperpolarizing shift in postspike voltage speeds deactivation of Kv3-like potassium channels. Other experiments examined bursting in Nav1.6-/- mice, in which sodium current density is reduced in the long term. Under these circumstances, there was upregulation of both T-type and P-type calcium current and a change in the balance of calcium current and

calcium-activated potassium current such that their net influence shifted from being inhibitory during bursts in wild-type neurons to excitatory during bursts from Nav1.6-/- mutant neurons. Thus, Purkinje neurons have both acute and long-term feedback mechanisms that serve to maintain burst firing when voltage-dependent sodium conductance is reduced."

Location and anatomical distribution of dlBNST/BNST CRF neurons

BNST CRF neurons are not spread evenly through the BNST. In mice they are concentrated in lateral dorsal compartments, especially through the middle anterior-posterior extent, and this lateral placement helps distinguish them from neighboring medial peptide-defined populations. (11 sources)

Anatomically, CRF-expressing BNST neurons are best placed within the dorsal/lateral part of the anterior BNST rather than treated as a diffuse BNST-wide pool. A useful summary is that **Crf** and **Cck** mark non-overlapping GABAergic populations that sit in separate **lateral versus medial subdivisions** of the middle BNST, spanning about **+0.2 to −0.2 mm from bregma** in mouse; in this scheme, CRF cells occupy the lateral compartment adjacent to medial CCK cells. (Giardino et al., 2021) (Giardino et al., 2018)

Within the broader dorsolateral BNST, there is further subregional structure. The **oval nucleus (ovBNST)** is a distinct subnucleus inside the larger dlBNST, and markers such as **Drd1** and **Pkcd** cluster there, while **calretinin** labeling is more selective for the **dlBNST** more generally. This means dlBNST CRF neurons sit in a molecularly patterned neighborhood rather than in a uniform field. (Giardino et al., 2021) (Kim et al., 2013) (Nguyen et al., 2016) (Wang et al., 2019)

More broadly, the BNST contains many other peptide-defined neurons intermingled across its subdivisions, including **Pdyn, Penk, Nts, Npy, Pnoc, Sst, Tac1, Tac2, and Avp** populations. So even where CRF neurons are enriched laterally, they are embedded in a dense patchwork of nearby cell classes rather than forming a sharply isolated anatomical island. (Giardino et al., 2021) (Ahrens et al., 2018) (Whylings et al., 2019) (Smith et al., 2019) (Zelikowsky et al., 2018) (Kovner et al., 2019)

This lateralized placement also fits with circuit anatomy. dlBNST neurons project strongly to targets such as the **lateral hypothalamus**, and CRF-defined BNST populations were identified among the LH-projecting GABAergic neurons, supporting the idea that dlBNST CRF cells are a spatially and projection-defined component of the dorsal BNST. (Giardino et al., 2021) (Giardino et al., 2018) (Barbier et al., 2020)

A final anatomical point is that “CRF BNST neurons” are not perfectly identical to a single transmitter-defined class. In the anterior BNST, CRF-expressing cells include both **GABAergic and glutamatergic subpopulations**, so their map in tissue is partly peptide-defined and partly transmitter-diverse. (Nguyen et al., 2016)

Evidence

(Giardino et al., 2021)

Extended Amygdala Neuropeptide Circuitry of Emotional Arousal: Waking Up on the Wrong Side of the Bed Nuclei of Stria Terminalis

"Crf and Cck are non-overlapping GABAergic subpopulations that occupy separate lateral vs. medial subdivisions, co-residing in adjacent compartments throughout the middle of the BNST A/P axis, approximately +0.2 to −0.2 mm from Bregma in the mouse. In addition to Crf and Cck, the neuropeptides dynorphin (Pdyn), enkephalin (Penk), neurotensin (Nts), neuropeptide Y (Npy), nociceptin (Pnoc), somatostatin (Sst), substance P (Tac1), neurokinin B (Tac2), and vasopressin (Avp) are also found in neurons throughout the BNST (Malsbury et al., 1987)(Walter et al., 1991)(Poulin et al., 2009)(Kudo et al., 2014)(Crowley et al., 2016)(Ahrens et al., 2018) (Giardino et al., 2018)(Zelikowsky et al., 2018)(Kovner et al., 2019)Rigney et al., 2019;(212758207)(Smith et al., 2019)(Whylings et al., 2019)(Xiao et al., 2020); Figure 1A). Labeling for the calcium-binding protein calretinin appears selective for the dorsolateral (dl)BNST, whereas dopamine receptor type-1 (Drd1) and protein kinase C delta (Pkcd) neurons cluster more specifically within the oval nucleus (ovBNST, a discrete subnucleus within the larger dlBNST subregion; (Kim et al., 2013)(Nguyen et al., 2016)(Pomrenze et al., 2019)(Wang et al., 2019)"

".Breakthrough efforts to characterize the entire genetic diversity of the BNST and surrounding regions at the single-cell level provided evidence for up to 37 distinct neuronal subtypes, although an exhaustive discussion of this data is beyond the scope of our review (Moffitt et al., 2018)(Welch et al., 2019). A more recent single-cell RNA sequencing study targeted specifically in the dorsal (d)BNST identified several neuronal clusters, including those marked by expected genes (e.g., Pkcd, Sst, Npy), but also some

surprising markers (e.g., Lmo4; Rodriguez-Romaguera et al., 2020)"

".(Barbier et al., 2020) traced the long-range projections of dlBNST and dorsomedial (dm)BNST neurons in exquisite detail, finding that the dlBNST projects especially to the LH and tuberomammillary nucleus (TMN; the site of wake-promoting histamine neurons; (Barbier et al., 2020)"

".Another study from the Kash group found a complex microcircuit in the BNST comprised of Crf interneurons that are modulated by serotonin and regulate the activity of separate long-range BNST projection neurons (Marcinkiewcz et al., 2016)"

".dlBNST Crf neurons preferentially respond to aversive stimuli and drive behavioral avoidance"

(Giardino et al., 2018)

PARALLEL CIRCUITS FROM THE BED NUCLEI OF STRIA TERMINALIS TO THE LATERAL HYPOTHALAMUS DRIVE OPPOSING EMOTIONAL STATES

"Lateral hypothalamus (LH) neurons containing the neuropeptide hypocretin (HCRT; orexin) modulate affective components of arousal, but their relevant synaptic inputs remain poorly defined. Here we identified inputs onto LH neurons that originate from neuronal populations in the bed nuclei of stria terminalis (BNST; a heterogeneous region of extended amygdala). We characterized two non-overlapping LH-projecting GABAergic BNST subpopulations that express distinct neuropeptides (corticotropin-releasing factor, CRF, and cholecystokinin, CCK). To functionally interrogate BNST LH circuitry, we used tools for monitoring and manipulating neural activity with cell-type-specific resolution in freely behaving mice. We found that Crf-BNST and Cck-BNST neurons respectively provide abundant and sparse inputs onto Hcrt-LH neurons, display discrete physiological responses to salient stimuli, drive opposite emotionally valenced behaviors, and receive different proportions of inputs from upstream networks. Together, our data provide an advanced model for how parallel BNST LH pathways promote divergent emotional states via connectivity patterns of genetically defined, circuit-specific neuronal subpopulations. Using genetically encoded physiological tools for monitoring, manipulating, and mapping discrete neural circuits, Giardino et al. characterize 2 discrete amygdala hypothalamus pathways that promote opposite behavioral responses to emotional stimuli."

(Kim et al., 2013)

Diverging neural pathways assemble a behavioural state from separable features in anxiety

"Behavioural states in mammals, such as the anxious state, are characterized by several features that are coordinately regulated by diverse nervous system outputs, ranging from behavioural choice patterns to changes in physiology (in anxiety, exemplified respectively by risk-avoidance and respiratory rate alterations). Here we investigate if and how defined neural projections arising from a single coordinating brain region in mice could mediate diverse features of anxiety. Integrating behavioural assays, in vivo and in vitro electrophysiology, respiratory physiology and optogenetics, we identify a surprising new role for the bed nucleus of the stria terminalis (BNST) in the coordinated modulation of diverse anxiety features. First, two BNST subregions were unexpectedly found to exert opposite effects on the anxious state: oval BNST activity promoted several independent anxious state features, whereas anterodorsal BNST-associated activity exerted anxiolytic influence for the same features. Notably, we found that three distinct anterodorsal BNST efferent projections—to the lateral hypothalamus, parabrachial nucleus and ventral tegmental area—each implemented an independent feature of anxiolysis: reduced risk-avoidance, reduced respiratory rate, and increased positive valence, respectively. Furthermore, selective inhibition of corresponding circuit elements in freely moving mice showed opposing behavioural effects compared with excitation, and in vivo recordings during free behaviour showed native spiking patterns in anterodorsal BNST neurons that differentiated safe and anxiogenic environments. These results demonstrate that distinct BNST subregions exert opposite effects in modulating anxiety, establish separable anxiolytic roles for different anterodorsal BNST projections, and illustrate circuit mechanisms underlying selection of features for the assembly of the anxious state."

(Nguyen et al., 2016)

Genetic cell targeting uncovers specific neuronal types and distinct subregions in the bed nucleus of the stria terminalis

"In addition, we discovered that corticotropin-releasing hormone-expressing cells contain GABAergic and glutamatergic subpopulations"

". Since considerable progress has been made in correlating cell types with chemical markers, particularly in the neocortex and hippocampus, it is suspected that some of neurochemical markers examined in the current study can be closely correlated with morphology and/or physiology of specific BNST neuron types"

". by using the new information derived from the present study, specific cell types can be targeted to map their circuit connections and assess their functional roles in BNST-associated behaviors."

(Wang et al., 2019)

A bed nucleus of stria terminalis microcircuit regulating inflammation-associated modulation of feeding

"Loss of appetite or anorexia associated with inflammation impairs quality of life and increases morbidity in many diseases. However, the exact neural mechanism that mediates inflammation-associated anorexia is still poorly understood. Here we identified

a population of neurons, marked by the expression of protein kinase C-delta, in the oval region of the bed nucleus of the stria terminalis (BNST), which are activated by various inflammatory signals. Silencing of these neurons attenuates the anorexia caused by these inflammatory signals. Our results demonstrate that these neurons mediate bidirectional control of general feeding behaviors. These neurons inhibit the lateral hypothalamus-projecting neurons in the ventrolateral part of BNST to regulate feeding, receive inputs from the canonical feeding regions of arcuate nucleus and parabrachial nucleus. Our data therefore define a BNST microcircuit that might coordinate canonical feeding centers to regulate food intake, which could offer therapeutic targets for feeding-related diseases such as anorexia and obesity. Inflammation can reduce food intake. Here the authors show that the GABAergic pathway from bed nucleus of stria terminalis to lateral hypothalamus regulates the inflammation induced reduction in feeding in mice."

(Ahrens et al., 2018)

A Central Extended Amygdala Circuit That Modulates Anxiety

"Both the amygdala and the bed nucleus of the stria terminalis (BNST) have been implicated in maladaptive anxiety characteristics of anxiety disorders. However, the underlying circuit and cellular mechanisms have remained elusive. Here we show that mice with Erbb4 gene deficiency in somatostatin-expressing (SOM+) neurons exhibit heightened anxiety as measured in the elevated plus maze test and the open field test, two assays commonly used to assess anxiety-related behaviors in rodents. Using a combination of electrophysiological, molecular, genetic, and pharmacological techniques, we demonstrate that the abnormal anxiety in the mutant mice is caused by enhanced excitatory synaptic inputs onto SOM+ neurons in the central amygdala (CeA), and the resulting reduction in inhibition onto downstream SOM+ neurons in the BNST. Notably, our results indicate that an increase in dynorphin signaling in SOM+ CeA neurons mediates the paradoxical reduction in inhibition onto SOM+ BNST neurons, and that the consequent enhanced activity of SOM+ BNST neurons is both necessary for and sufficient to drive the elevated anxiety. Finally, we show that the elevated anxiety and the associated synaptic dysfunctions and increased dynorphin signaling in the CeA–BNST circuit of the Erbb4 mutant mice can be recapitulated by stress in wild-type mice. Together, our results unravel previously unknown circuit and cellular processes in the central extended amygdala that can cause maladaptive anxiety. SIGNIFICANCE STATEMENT The central extended amygdala has been implicated in anxiety-related behaviors, but the underlying mechanisms are unclear. Here we found that somatostatin-expressing neurons in the central amygdala (CeA) controls anxiety through modulation of the stria terminalis, a process that is mediated by an increase in dynorphin signaling in the CeA. Our results reveal circuit and cellular dysfunctions that may account for maladaptive anxiety."

(Whylings et al., 2019)

Sexually dimorphic role of BNST vasopressin cells in sickness and social behavior in male and female mice

"Circumstantial evidence supports the hypothesis that the sexually dimorphic vasopressin (AVP) innervation of the brain tempers sickness behavior in males. Here we test this hypothesis directly, by comparing sickness behavior in animals with or without ablations of BNST AVP cells, a major source of sexually dimorphic AVP in the brain. We treated male and female AVP-iCre+ and AVP-iCre− mice that had been injected with viral Cre-dependent caspase-3 executioner construct into the BNST with lipopolysaccharide (LPS) or sterile saline, followed by behavioral analysis. In all groups, LPS treatment reliably reduced motor behavior, increased anxiety-related behavior, and reduced sucrose preference and consumption. Male mice, whose BNST AVP cells had been ablated (AVP-iCre+), displayed only minor reductions in LPS-induced sickness behavior, whereas their female counterparts displayed, if anything, an increase in sickness behaviors. All saline-treated mice with BNST AVP cell ablations consumed more sucrose than did control mice, and males, but not females, with BNST AVP cell ablations showed reduced preference for novel conspecifics compared to control mice. These data confirm that BNST AVP cells control social behavior in a sexually dimorphic way, but do not play a critical role in altering sickness behavior."

(Smith et al., 2019)

Extrahypothalamic GABAergic nociceptin–expressing neurons regulate AgRP neuron activity to control feeding behavior

"Arcuate nucleus agouti–related peptide (AgRP) neurons play a central role in feeding and are under complex regulation by both homeostatic hormonal and nutrient signals and hypothalamic neuronal pathways. Feeding may also be influenced by environmental cues, sensory inputs, and other behaviors, implying the involvement of higher brain regions. However, whether such pathways modulate feeding through direct synaptic control of AgRP neuron activity is unknown. Here, we show that nociceptin-expressing neurons in the anterior bed nuclei of the stria terminalis (aBNST) make direct GABAergic inputs onto AgRP neurons. We found that activation of these neurons inhibited AgRP neurons and feeding. The activity of these neurons increased upon food availability, and their ablation resulted in obesity. Furthermore, these neurons received afferent inputs from a range of upstream brain regions as well as hypothalamic nuclei. Therefore, aBNST GABAergic nociceptin neurons may act as a gateway to feeding behavior by connecting AgRP neurons to both homeostatic and nonhomeostatic neuronal inputs."

(Zelikowsky et al., 2018)

The neuropeptide Tac2 controls a distributed brain state induced by chronic social isolation stress

"Chronic social isolation causes severe psychological effects in humans, but their neural bases remains poorly understood. Two weeks (but not 24 hrs) of social isolation stress (SIS) caused multiple behavioral changes in mice, and induced brain-wide up-regulation of the neuropeptide tachykinin 2 (Tac2)/neurokinin B (NkB). Systemic administration of an Nk3R antagonist prevented

virtually all of the behavioral effects of chronic SIS. Conversely, enhancing NkB expression and release phenocopied SIS in group-housed mice, promoting aggression and converting stimulus-locked defensive behaviors to persistent responses. Multiplexed analysis of Tac2/NkB function in multiple brain areas revealed dissociable, region-specific requirements for both the peptide and its receptor in different SIS-induced behavioral changes. Thus, Tac2 coordinates a pleiotropic brain state caused by SIS, via a distributed mode of action. These data reveal the profound effects of prolonged social isolation on brain chemistry and function, and suggest potential new therapeutic applications for Nk3R antagonists."

(Kovner et al., 2019)

Somatostatin Gene and Protein Expression in the Non-human Primate Central Extended Amygdala

"Alterations in central extended amygdala (EAc) function have been linked to anxiety, depression, and anxious temperament (AT), the early-life risk to develop these disorders. The EAc is composed of the central nucleus of the amygdala (Ce), the bed nucleus of the stria terminalis (BST), and the sublenticular extended amygdala (SLEA). Using a non-human primate model of AT and multimodal neuroimaging, the Ce and the BST were identified as key AT-related regions. Both areas are primarily comprised of GABAergic neurons and the lateral Ce (CeL) and lateral BST (BSTL) have among the highest expression of neuropeptides in the brain. Somatostatin (SST) is of particular interest because mouse studies demonstrate that SST neurons, along with corticotropin releasing factor (CRF) neurons, contribute to a threat-relevant EAc microcircuit. Although the distribution of CeL and BSTL SST neurons has been explored in rodents, this system is not well described in non-human primates. In situ hybridization demonstrated an anterior-posterior gradient of SST mRNA in the CeL but not the BSTL of non-human primates. Triple labeling immunofluorescence staining revealed that SST protein expressing cell bodies are a small proportion of the total CeL and BSTL neurons and have considerable co-labeling with CRF. The SLEA exhibited strong SST mRNA and protein expression, suggesting a role for SST in mediating information transfer between the CeL and BSTL. These data provide the foundation for mechanistic non-human primate studies focused on understanding EAc function in neuropsychiatric disorders."

(Barbier et al., 2020)

Projections from the dorsomedial division of the bed nucleus of the stria terminalis to hypothalamic nuclei in the mouse

"As stressful environment is a potent modulator of feeding, we seek in the present work to decipher the neuroanatomical basis for an interplay between stress and feeding behaviors. For this, we combined anterograde and retrograde tracing with immunohistochemical approaches to investigate the patterns of projections between the dorsomedial division of the bed nucleus of the stria terminalis (BNST), well connected to the amygdala, and hypothalamic structures such as the paraventricular (PVH) and dorsomedial (DMH), the arcuate (ARH) nuclei and the lateral hypothalamic areas (LHA) known to control feeding and motivated behaviors. We particularly focused our study on afferences to proopiomelanocortin (POMC), agouti-related peptide (AgRP), melanin-concentrating-hormone (MCH) and orexin (ORX) neurons characteristics of the ARH and the LHA, respectively. We found light to intense innervation of all these hypothalamic nuclei. We particularly showed an innervation of POMC, AgRP, MCH and ORX neurons by the dorsomedial and dorsolateral divisions of the BNST. Therefore, these results lay the foundation for a better understanding of the neuroanatomical basis of the stress-related feeding behaviors."

Molecular identity and markers

*dlBNST/BNST CRF neurons are best defined first by **Crh/CRF** expression, but they are not chemically uniform. Most evidence points to a mainly **GABAergic** identity in the dlBNST/BNSTALG, with smaller **glutamatergic** CRF subsets elsewhere in anterior BNST, plus distinctive receptor and ion-channel marker patterns. (5 sources)*

- **Core defining marker: CRF/Crh**
- The defining feature of this population is expression of **corticotropin-releasing factor (CRF; Crh)**, which has been used genetically and anatomically to target BNST neurons tied to stress and negative emotional states. (Salimando et al., 2020)
- **Transmitter phenotype is mostly GABAergic in dlBNST, but not exclusively so across BNST**
- A key molecular point is that BNST CRF neurons are **not a purely single-transmitter class** across the whole anterior BNST: CRF-expressing cells include both **GABAergic and glutamatergic subpopulations**. (Nguyen et al., 2016)
- At the same time, work focused on the BNST anterolateral group/dlBNST strongly supports a **predominantly GABAergic phenotype** for this local CRF population. In BNSTALG, all tested neurons expressed **GAD67** transcripts, and CRF-rich Type III cells lacked **VGLUT1-3** transcripts. (Dabrowska et al., 2013)

- This regional bias is also reflected in later summaries of the dlBNST literature, which note that about **98% of CRF+ neurons in dlBNST are GABAergic**. (Companion et al., 2018) (Nguyen et al., 2016)
- **Association with a physiological cell class**
- In the BNSTALG, CRF identity maps strongly onto the classic **Type III** physiological class: about **95% of Type III neurons express CRF mRNA**, linking the peptide marker to a broader membrane-property and ion-channel profile rather than to peptide expression alone. (Dabrowska et al., 2013) (Hazra et al., 2011)
- **Receptor markers that distinguish BNST CRF neurons from hypothalamic CRF neurons**
- BNST Type III/CRF neurons differ from paraventricular hypothalamic CRF neurons in peptide-receptor makeup. A notable distinction is that **95% of Type III neurons expressed oxytocin receptor (OTR) transcripts**, whereas parvocellular PVN CRF neurons did not; conversely, BNST Type III CRF neurons did **not** co-express **oxytocin (OT)** itself. (Dabrowska et al., 2013)
- These marker differences support the idea that BNST CRF neurons are a separate extra-hypothalamic CRF cell class, not just a displaced version of PVN CRF neurons. (Dabrowska et al., 2013)
- **Glutamate receptor expression**
- dlBNST CRF neurons can also be marked functionally by expression of the NMDA receptor subunit **GluN2D**, which was reported to be functionally expressed on BNST CRF-positive cells involved in negative emotional states. (Salimando et al., 2020)
- **Ion-channel transcript signatures**
- Molecular profiling of BNSTALG neurons showed that physiological classes differ in transcripts for ion-channel alpha subunits, including **HCN1-4, Cav3.1-3.3, Kv1.4, Kv3.4, Kv4.1-4.3, and Kir2.1-2.4**. This does not by itself define CRF neurons, but it gives a marker framework for the Type III/CRF-linked class and suggests that dlBNST CRF neurons can be recognized by combined peptide plus ion-channel signatures rather than by CRF alone. (Hazra et al., 2011) (Dabrowska et al., 2013)
- **Take-home marker set**
- A practical marker panel for dlBNST CRF neurons from these studies would include **CRF/Crh**, **GAD67/GABAergic identity** as the main transmitter label, **absence of VGLUT1-3** for the classic BNSTALG/dlBNST CRF phenotype, frequent **OTR** expression, and in some contexts **GluN2D** as a functionally relevant receptor marker. (Dabrowska et al., 2013) (Salimando et al., 2020) (Nguyen et al., 2016)

Paper	Neuronal Types Identified	Techniques Used	Molecular Markers Used	Neurotransmitter Phenotype	Functional Role In Behavior	Neuronal Subpopulations Or Subtypes

Paper	Neuronal Types Identified	Techniques Used	Molecular Markers Used	Neurotransmitter Phenotype	Functional Role In Behavior	Neuronal Subpopulations Or Subtypes
BNST GluN2D-Containing NMDA Receptors Influence Anxiety- and Depressive-like Behaviors and ModulateCell-Specific Excitatory/Inhibitory Synaptic Balance	CRF-positive BNST cells and GluN2D-expressing neurons.	Ex vivo electrophysiology, in vivo fiber photometry, GluN2D KO mouse models.	GluN2D knockout mouse lines (GluN2D−/−, GluN2Dflx/flx); CRF-positive BNST cells.	Corticotropin-releasing factor (CRF)-positive neurons, increased excitatory drive.	GluN2D-containing NMDARs modulate anxiety, depressive-like behaviors, and BNST synaptic activity.	GluN2D expressed on corticotropin-releasing factor (CRF)-positive BNST cells.
Genetic cell targeting uncovers specific neuronal types and distinct subregions in the bed nucleus of the stria terminalis	Excitatory (glutamatergic) and inhibitory (GABAergic) neurons; somatostatin-, calretinin-, parvalbumin-expressing neurons.	Genetic cell targeting, immunochemical staining, mouse genetics, Cre driver lines.	Cre driver lines, fluorescent reporter line, somatostatin, calretinin, vasoactive intestinal peptide, corticotropin-releasing hormone.	Glutamatergic neurons: 14% dorsal, 31% ventral BNST; GABAergic neurons: differential subregional distribution.	N/A	Specific excitatory and inhibitory BNST cell types identified, GABAergic and glutamatergic subpopulations detailed.

Paper	Neuronal Types Identified	Techniques Used	Molecular Markers Used	Neurotransmitter Phenotype	Functional Role In Behavior	Neuronal Subpopulations Or Subtypes
Central CRF neurons are not created equal: phenotypic differences in CRF-containing neurons of the rat paraventricular hypothalamus and the bed nucleus of the stria terminalis	BNST CRF neurons are GABAergic; PVN CRF neurons are glutamatergic.	Dual-immunofluorescence, single-cell RT-PCR, electrophysiological techniques.	CRF, oxytocin, arginine-vasopressin expression	PVN CRF neurons are glutamatergic; BNST CRF neurons are GABAergic.	CRF mediates stress and anxiety responses; BNST dysfunction linked to anxiety, depression.	PVN CRF neurons are glutamatergic; BNST CRF neurons are GABAergic.
Assessment of ventral tegmental area-projecting GABAergic neurons from the bed nucleus of the stria terminalis in modulating binge-like ethanol intake	CRF-expressing GABAergic neurons within the dlBNST.	Chemogenetics, immunohistochemistry, viral vectors, clozapine-N-oxide infusion.	Vgat-ires-Cre transgenic mice, CRF-expressing neurons.	CRF-expressing GABAergic neurons in dlBNST modulate EtOH consumption.	VTA-projecting dlBNST GABAergic neurons reduce binge-like ethanol consumption.	VTA-projecting dlBNST GABAergic neurons, CRF-expressing within dlBNST.

Paper	Neuronal Types Identified	Techniques Used	Molecular Markers Used	Neurotransmitter Phenotype	Functional Role In Behavior	Neuronal Subpopulations Or Subtypes
A transcriptomic analysis of type I-III neurons in the bed nucleus of the stria terminalis	Type I, Type II, Type III neurons; Type II further subdivided.	Whole cell patch clamp recording, single cell reverse transcriptase polymerase chain reaction, hierarchical cluster analysis, discriminant function analysis.	Expression of α subunits for Ih, IT, IA, and IK(IR) ion channels.	N/A	Activity of BNSTALG neurons is critical in anxiety- and stress-related behaviors.	Type I-III neurons, Type II subdivided into three genetic subgroups.

Evidence

(Salimando et al., 2020)

BNST GluN2D-Containing NMDA Receptors Influence Anxiety- and Depressive-like Behaviors and ModulateCell-Specific Excitatory/ Inhibitory Synaptic Balance

"We focused in particular on those cells expressing the stress-related peptide CRF, as this population has been previously implicated in the regulation of emotional"

"We also found that GluN2D is functionally expressed on corticotropin-releasing factor (CRF)-positive BNST cells implicated in driving negative emotional states"

"To generate the necessary transgenic animals, we crossed our GluN2D À/À mice to the an Crh-IRES-Cre line, then injected Cre-dependent virus transducing the expression of the calcium-sensor GCaMP7f under control of the hSyn promoter unilaterally into the dlBNST"

"It is important to note that we cannot rule out that regional deletion of GluN2D in the BNST also affects CRFnegative cells, and may result in potential changes in the surrounding BNST microcircuitry that could factor into the control of excitatory/inhibitory tone at BNST-CRF cells (Gracy et al., 1995)(Paquet et al., 2000)(Turesson et al., 2013). Thus, further cell-specific manipulations of GluN2D-NMDAR function in the BNST are required to gain a more complete mechanistic understanding of its role in mediating regional excitatory signaling. Apart from extending these studies to the CRF-negative BNST cells that also express functional GluN2D-NMDARs, determining whether several of the key regions providing glutamatergic inputs to the BNST express the receptor (somatically, axonally, or dendritically) will also be essential to providing a more definitive understanding of GluN2D-mediated signaling in relation to stress-responsive or negative emotional behavior."

(Nguyen et al., 2016)

Genetic cell targeting uncovers specific neuronal types and distinct subregions in the bed nucleus of the stria terminalis

"In addition, we discovered that corticotropin-releasing hormone-expressing cells contain GABAergic and glutamatergic subpopulations"

". Since considerable progress has been made in correlating cell types with chemical markers, particularly in the neocortex and hippocampus, it is suspected that some of neurochemical markers examined in the current study can be closely correlated with morphology and/or physiology of specific BNST neuron types"

". by using the new information derived from the present study, specific cell types can be targeted to map their circuit connections

and assess their functional roles in BNST-associated behaviors."

(Dabrowska et al., 2013)

Central CRF neurons are not created equal: phenotypic differences in CRF-containing neurons of the rat paraventricular hypothalamus and the bed nucleus of the stria terminalis

"Previously, we have shown that three physiologically and genetically distinct subclasses of neurons are present in the BNST ALG , Type I-III (Hammack et al., 2007;(Hazra et al., 2011). Type III BNST ALG neurons are putative CRF neurons, since the majority (95%) of these neurons express mRNA transcripts for CRF, and display a unique ion channel expression pattern that correlates with their electrophysiological features (Hazra et al., 2011). To determine if CRF neurons of the BNST ALG share a phenotype with parvocellular CRF neurons of the PVN we screened Type I-III BNST neurons for the presence of GAD67 and VGLUT1-3 transcripts. Significantly, all BNST ALG neurons tested, irrespective of cell type, expressed GAD67 mRNA transcripts, but none of these neurons expressed transcripts for VGLUT1-3. These findings confirmed our immunohistochemical observations that CRF neurons in the BNST ALG have a predominantly GABAergic phenotype"

". Significantly, unlike parvocellular CRF neurons that showed 83% co-expression with OT mRNA transcripts, none of the Type III CRF neurons tested co-expressed OT. Conversely, whereas 95% of Type III neurons expressed transcripts for OTR (18/19), no parvocellular CRF neurons expressed OTR transcripts."

(Companion et al., 2018)

Assessment of ventral tegmental area-projecting GABAergic neurons from the bed nucleus of the stria terminalis in modulating binge-like ethanol intake

"These CRF projection neurons have been shown to exert a GABAergic phenotype (Dabrowska et al., 2013)"

".We then applied immunohistochemistry (IHC) to assess co-expression of CRF with our DREADD virus in dlBNST neurons"

".A qualitative assessment of the co-expression suggests that most, if not all, of the AAV8-hSyn-DIO-hM4d-mCherry+ neurons also expressed CRF immunoreactivity, consistent with the observation that 98% of GABAergic neurons in the dlBNST entail a CRF phenotype (Nguyen et al., 2016)"

".A recent study found that while CRF+ neurons in the BNST co-express both glutamate and GABA, within the dlBNST, about 98% of CRF+ neurons are GABAergic (Nguyen et al., 2016)."

(Hazra et al., 2011)

A transcriptomic analysis of type I-III neurons in the bed nucleus of the stria terminalis

"The activity of neurons in the anterolateral cell group of the bed nucleus of the stria terminalis (BNSTALG) plays a critical role in anxiety- and stress-related behaviors. Histochemical studies have suggested that multiple distinct neuronal phenotypes exist in the BNSTALG. Consistent with this observation, the physiological properties of BNSTALG neurons are also heterogeneous, and three distinct cell types can be defined (Type I–III) based primarily on their expression of four key membrane currents, namely Ih, IA, IT, and IK(IR). Significantly, all four channels are multimeric proteins and can comprise of more than one pore-forming α subunit. Hence, differential expression of α subunits may further diversify the neuronal population. However, nothing is known about the relative expression of these ion channel α subunits in BNSTALG neurons. We have addressed this lacuna by combining whole cell patch clamp recording together with single cell reverse transcriptase polymerase chain reaction (scRT-PCR) to assess the mRNA transcript expression for each of the subunits for the four key ion channels in Type I-III neurons of the BNSTALG. Here, cytosolic mRNA from single neurons was probed for the expression of transcripts for each of the α subunits of Ih (HCN1- HCN4), IT (Cav3.1- Cav3.3), IA (Kv1.4, Kv3.4, Kv4.1- Kv 4.3) and IK(IR) (Kir2.1-Kir2.4). An unbiased hierarchical cluster analysis followed by discriminant function analysis revealed that a positive correlation exists between the physiological and genetic phenotype of BNSTALG neurons. Thus, the analysis segregated BNSTALG neurons into 3 distinct groups, based on their α subunit mRNA expression profile, which positively correlated with our existing electrophysiological classification (Type I–III). Furthermore, analysis of mRNA transcript expression in Type I –Type III neurons suggested that, whereas Type I and III neurons appear to represent genetically homologous cell populations, Type II neurons may be further subdivided into three genetically distinct subgroups. These data not only validate our original classification scheme, but further refine the classification at the molecular level, and thus identifies novel targets for potential disruption and/or pharmacotherapeutic intervention in stress-related anxiety-like behaviors."

Physiological subtypes and heterogeneity

dlBNST/BNST CRF neurons are not one uniform physiological class. Even within CRF-linked and dlBNST-defined groups, recordings reveal multiple electrophysiological subtypes and state-dependent variation, with especially clear heterogeneity in type II neurons. (2 sources)

Physiological work argues strongly against treating BNST CRF neurons as a single membrane-property class. In mouse BNST, recordings from identified CRF neurons found **at least four electrophysiologically separable**

CRF-neuron subtypes, a level of diversity greater than older, simpler schemes suggested and notably greater than the relative homogeneity seen for CeA CRF neurons recorded under similar conditions. The same study also noted that some of the apparent subtype structure could reflect different activation states, but taken together with prior molecular work it supports the idea that BNST CRF neurons contain real within-class physiological diversity rather than mere recording noise. This is one reason projection labels or molecular markers are useful for parsing BNST CRF subpopulations more precisely. (Silberman et al., 2013)

This heterogeneity is not limited to CRF-positive cells narrowly defined. In the dlBNST, **type II neurons** show especially clear internal diversity: only type II neurons responded to both **CRF and NPY**, and current-voltage analyses suggested **at least two classes of type II dlBNST neurons**. In other words, even one electrophysiological "type" can break into smaller response-defined subclasses. (Ide et al., 2013)

Taken together with the earlier point that CRF identity in BNST often aligns with, but is not exhausted by, classic physiological typing, the practical message is that dlBNST CRF neurons are best viewed as a **heterogeneous ensemble**. Some differences likely map onto stable subtype features, while others may depend on circuit state, peptide sensitivity, or projection target. (Silberman et al., 2013) (Ide et al., 2013)

Evidence

(Silberman et al., 2013)

A Corticotropin Releasing Factor Pathway for Ethanol Regulation of the Ventral Tegmental Area in the Bed Nucleus of the Stria Terminalis

> *"Our data indicate that there are at least four electrophysiologically separable subtypes of CRF neurons and three types of VTA-projecting neurons in the mouse BNST. The presence of at least four categories of electrophysiologically distinct categories of BNST neurons is more than previously described and could be due to differences in rat and mouse BNST physiology, differences in recording sites within the slice, or differences in recording conditions. However, CeA CRF neurons seem much more homogenous in nature than BNST CRF neurons recorded under similar conditions, suggesting that our recording conditions are not a major factor in BNST neuron subtype variability. Indeed, differences between BNST neuronal subtypes may to some degree be due to one neuronal subtype recorded during different activation states, although evidence does suggest unique mRNA profiles (Hammack et al., 2007). These data point to the use of fluorescent markers based on neuronal protein content or terminal projections as an important approach to study specific BNST neuron subpopulations."*

(Ide et al., 2013)

Opposing Roles of Corticotropin-Releasing Factor and Neuropeptide Y within the Dorsolateral Bed Nucleus of the Stria Terminalis in the Negative Affective Component of Pain in Rats

> *"Interestingly, we found that only type II, not type I or III, neurons responded to both CRF and NPY in the dlBNST"*
>
> *". Analyses of I-V relationships in the absence and presence of CRF demonstrate that there may be at least two classes of type II dlBNST neurons"*
>
> *". In this study, we revealed the heterogeneity of type II dlBNST neurons."*

Connectivity and circuit organization

dlBNST/BNST CRF neurons sit in a circuit node with both local and long-range links. The best-supported motifs are local CRF signaling within dlBNST, CRF input from CeA/CeL to BNSTDL, and BNST CRF outputs to hypothalamic and reward-related targets such as LH, PVH, and VTA-linked pathways. (10 sources)

- **Local dlBNST CRF microcircuitry is likely important**
- Evidence supports the idea that at least some dlBNST CRF signaling comes from **intrinsic BNST neurons**, not only from afferents. Work on dlBNST physiology proposed that some **type II neurons may act as local-circuit GABAergic interneurons** that contact other dlBNST neuron types, and the same line of work identified intrinsic dlBNST neurons as one likely source of CRF terminals in this region. (Takahashi et al., 2019) (Silberman et al., 2013) (Pomrenze et al., 2015)
- This local arrangement fits physiological data showing that CRF can **excite type II dlBNST neurons** while

also being linked to **increased inhibitory synaptic input onto type III neurons**, suggesting that CRF neurons can shape dlBNST output indirectly through intra-BNST interactions. (Takahashi et al., 2019) (Ide et al., 2013)

- **A major extrinsic CRF input to BNSTDL/dlBNST comes from the central amygdala**
- A well-supported pathway runs from **CeA/CeL CRF neurons to dorsal and ventral BNST**, including the dlBNST/BNSTDL domain. Viral tracing found major CeA CRF projections to BNST, and fear-circuit work showed that **CeAL CRF axons cluster around BNSTDL CRF cells**. (Pomrenze et al., 2015) (Asok et al., 2017)
- Functionally, this CeAL→BNSTDL CRF pathway contributes to communication between amygdala and BNST during fear learning, supporting the idea that dlBNST CRF neurons are embedded in a **CRF-to-CRF extended amygdala channel** rather than acting in isolation. (Asok et al., 2017)
- **CRF in dlBNST can come from both local and CeA sources**
- Studies focused on pain-related plasticity explicitly identify **two candidate sources of CRF-containing terminals in dlBNST**: **intrinsic dlBNST neurons** and **CeA neurons that project to dlBNST**. In chronic pain models, **CRF mRNA increased in both dlBNST and CeA**, consistent with recruitment of both arms of this circuit. (Takahashi et al., 2019) (Arakaki et al., 2021)
- **BNST CRF neurons are part of a notable BNST→LH projection system**
- A notable projection-defined CRF population in BNST sends axons to the **lateral hypothalamus (LH)**. In mice, **Crf-BNST** and **Cck-BNST** neurons form non-overlapping LH-projecting GABAergic channels with different input patterns and opposite behavioral effects; importantly, **Crf-BNST neurons provide abundant input onto orexin/hypocretin LH neurons**. (Giardino et al., 2018)
- More broadly, tracing studies also identify **LH** as a recurrent target of BNST CRF neurons. (Romero et al., 2026) (Pomrenze et al., 2015)
- **BNST CRF neurons also target hypothalamic stress-output regions**
- Anatomical tracing has reported BNST CRF projections to the **paraventricular hypothalamic nucleus (PVH)** as well as LH, placing these cells in position to influence endocrine and autonomic stress pathways. (Romero et al., 2026)
- **CRF signaling in BNST can regulate VTA-projecting BNST neurons**
- Even when the recorded cells are not themselves CRF-positive, BNST CRF neurons appear to modulate a **BNST→VTA output channel**. Dopamine and β-adrenergic stimulation depolarize BNST CRF neurons, and CRF increases excitatory synaptic drive onto **VTA-projecting BNST neurons**, supporting a local CRF relay from catecholamine-sensitive BNST CRF cells to projection neurons. (Silberman et al., 2013)
- In chronic pain work, **VTA-projecting dlBNST neurons** were reported to receive increased inhibitory input mediated by enhanced CRF signaling, and these projection neurons can synapse onto **VTA GABAergic neurons**. (Arakaki et al., 2021)
- **dlBNST/BNST CRF neurons should be placed within broader BNST input-output architecture**
- The broader BNST, especially anterodorsal/lateral sectors related to dlBNST territory, receives strong afferent input from **amygdala, hypothalamus, and hippocampal formation**, with lateral parts receiving relatively more **prefrontal, insular, thalamic, and perirhinal/ectorhinal** input. This gives a circuit context for why dlBNST CRF neurons can integrate threat, interoceptive, and motivational signals. (Sun et al., 2023)
- At the output side, BNST projections broadly reach **hypothalamus, brainstem, and reward/motivation circuits**. Classic projection work further showed that different BNST efferents can control different anxiety-

related features, including **BNST→LH**, **BNST→PBN**, and **BNST→VTA** channels. (Kim et al., 2013) (Romero et al., 2026)

- **Working circuit picture**
- A compact way to think about dlBNST CRF neurons is that they are positioned at the intersection of **CeA-derived CRF input**, **local BNST CRF microcircuits**, and **downstream hypothalamic and mesolimbic outputs**. This organization gives them a plausible role in converting stress-related input into changes in local inhibition and long-range control of arousal, avoidance, motivation, and autonomic/endocrine state. (Pomrenze et al., 2015) (Asok et al., 2017) (Giardino et al., 2018) (Silberman et al., 2013)

Paper	Focus On Crf Pathways	Functional Outcomes Studied	Neuronal Activity Analysis Techniques	Connectivity With Other Brain Regions	Diversity And Subpopulations Of Bnst Neurons	Techniques For Structural/ Functional Mapping
Tonic Suppression of the Mesolimbic Dopaminergic System by Enhanced Corticotropin-Releasing Factor Signaling Within the Bed Nucleus of the Stria Terminalis in Chronic Pain Model Rats	CRF increased sIPSCs frequency in dlBNST; antagonist NBI27914 decreased sIPSCs in chronic pain rats.	Comorbidity of chronic pain and depression; conditioned place preference.	Whole-cell patch-clamp electrophysiology, sIPSCs measurement, brain slice analysis.	Increased sIPSCs in VTA-projecting dlBNST neurons indicate enhanced inhibitory connectivity to VTA in chronic pain.	N/A	Whole-cell patch-clamp electrophysiology, histological analyses.
A Corticotropin Releasing Factor Pathway for Ethanol Regulation of the Ventral Tegmental Area in the Bed Nucleus of the Stria Terminalis	CRF pathways assessed via neuronal activity, synaptic drive, and ethanol effects in BNST.	Stress- or cue-induced drug- and alcohol-seeking behaviors.	Whole-cell patch-clamp analysis, fluorescent reporter strategy, fluorescent microsphere-based identification.	BNST projections to VTA; CRF enhances synaptic drive onto VTA-projecting BNST neurons.	CRF neurons identified; roles in catecholamine interaction and excitatory drive onto VTA-projecting neurons.	CRF neuron fluorescent reporter, whole-cell patch-clamp, fluorescent microsphere-based identification.

Paper	Focus On Crf Pathways	Functional Outcomes Studied	Neuronal Activity Analysis Techniques	Connectivity With Other Brain Regions	Diversity And Subpopulations Of Bnst Neurons	Techniques For Structural/ Functional Mapping
A Transgenic Rat for Investigating the Anatomy and Function of Corticotrophin Releasing Factor Circuits	Projections from CeA CRF neurons to BNST and other brain regions.	GABA-ergic and excitatory CRF signaling, Fos induction in neurons.	Optogenetic and chemogenetic stimulation, Fos induction analysis.	Projections from CeA CRF neurons to BNST, VTA, and hypothalamus.	CeA CRF neuron projections to oval nucleus of BNST.	BAC transgenic Crh-Cre rat, Cre-dependent reporters, optogenetic and chemogenetic stimulation.
Histological analyses of CRF mRNA expression in the dorsolateral bed nucleus of the stria terminalis of chronic pain model rats	Enhanced CRF signaling in dlBNST and CeA via ISH and in vivo microdialysis.	Suppression of VTA dopaminergic neuron activity during chronic pain.	Electrophysiological analyses, in situ hybridization histochemistry, invivomicrodialysis.	VTA-projecting dlBNST neurons connect to VTA GABAergic neurons; CeA projects to dlBNST.	Increased inhibitory inputs to VTA-projecting dlBNST neurons via enhanced CRF signaling.	In situ hybridization histochemistry (ISH) and in vivo microdialysis.
PARALLEL CIRCUITS FROM THE BED NUCLEI OF STRIA TERMINALIS TO THE LATERAL HYPOTHALAMUS DRIVE OPPOSING EMOTIONAL STATES	Crf-BNST neurons provide abundant inputs to Hcrt-LH neurons; drive emotionally valenced behaviors.	Opposite emotionally valenced behaviors driven by Crf-BNST neuron inputs.	Monitoring and manipulating neural activity with cell-type-specific resolution in freely behaving mice.	Identified BNST LH pathways, distinct BNST subpopulations with different inputs.	Two GABAergic BNST subpopulations identified: Crf-BNST (abundant) and Cck-BNST (sparse), drive opposite emotional behaviors.	Cell-type-specific neural activity monitoring and manipulation tools in freely behaving mice.

Paper	Focus On Crf Pathways	Functional Outcomes Studied	Neuronal Activity Analysis Techniques	Connectivity With Other Brain Regions	Diversity And Subpopulations Of Bnst Neurons	Techniques For Structural/ Functional Mapping
Corticotropin-releasing factor neurons in the bed nucleus of the stria terminalis modulate avoidance behaviors and feeding	Fiber photometry, anatomical tracing, optogenetic stimulation in BNST CRF projections; LH and PVH pathways.	Avoidance behavior, feeding suppression.	Fiber photometry, axon terminal optogenetic stimulation.	BNST CRF neurons project to hypothalamic regions, including LH and PVH.	BNST CRF neurons project to LH and PVH; modulate avoidance and feeding.	Fiber photometry, anatomical tracing, axon terminal optogenetic stimulation.

Evidence

(Takahashi et al., 2019)

Tonic Suppression of the Mesolimbic Dopaminergic System by Enhanced Corticotropin-Releasing Factor Signaling Within the Bed Nucleus of the Stria Terminalis in Chronic Pain Model Rats

> *"Hammack et al. (2007) raised the possibility that some portion of type II dlBNST neurons may be local-circuit GABAergic interneurons, and that type II neurons should make synaptic contact with other type(s) of dlBNST neurons to regulate dlBNST output"*
>
> *". We previously reported that CRF selectively depolarizes dlBNST type II neurons (Ide et al., 2013) and increases the inhibitory synaptic inputs to dlBNST type III neurons (Nagano et al., 2015) in naive rats"*
>
> *". Two brain regions are considered possible sources of CRF-containing nerve terminals in the dlBNST: the dlBNST intrinsic neurons (Morin et al., 1999)(Silberman et al., 2013)(Pomrenze et al., 2015) and the CeA neurons projecting to the dlBNST (Asok et al., 2017). Histological analyses using a FISH technique revealed that CRF mRNA expression increased in the dlBNST and CeA of chronic pain model rats."*

(Silberman et al., 2013)

A Corticotropin Releasing Factor Pathway for Ethanol Regulation of the Ventral Tegmental Area in the Bed Nucleus of the Stria Terminalis

> *"Our data indicate that there are at least four electrophysiologically separable subtypes of CRF neurons and three types of VTA-projecting neurons in the mouse BNST. The presence of at least four categories of electrophysiologically distinct categories of BNST neurons is more than previously described and could be due to differences in rat and mouse BNST physiology, differences in recording sites within the slice, or differences in recording conditions. However, CeA CRF neurons seem much more homogenous in nature than BNST CRF neurons recorded under similar conditions, suggesting that our recording conditions are not a major factor in BNST neuron subtype variability. Indeed, differences between BNST neuronal subtypes may to some degree be due to one neuronal subtype recorded during different activation states, although evidence does suggest unique mRNA profiles (Hammack et al., 2007). These data point to the use of fluorescent markers based on neuronal protein content or terminal projections as an important approach to study specific BNST neuron subpopulations."*

(Pomrenze et al., 2015)

A Transgenic Rat for Investigating the Anatomy and Function of Corticotrophin Releasing Factor Circuits

> *"Using viral delivery of Cre-dependent reporters to identify CeL CRF neurons, we found robust CRF projections from CeL to the brainstem, terminating in the medial and lateral PBN and the LC (Figure 8). There were also extensive projections to the diencephalon, terminating in the dorsal and ventral BNST and the LH. This pattern of connectivity concurs with previously reported CeA CRF projections in the rat (Moga et al., 1985)(Sakanaka et al., 1986)(Bockstaele et al., 1998)(Reyes et al., 2011)."*

(Ide et al., 2013)

Opposing Roles of Corticotropin-Releasing Factor and Neuropeptide Y within the Dorsolateral Bed Nucleus of the Stria Terminalis

in the Negative Affective Component of Pain in Rats

"Interestingly, we found that only type II, not type I or III, neurons responded to both CRF and NPY in the dlBNST"

". Analyses of I-V relationships in the absence and presence of CRF demonstrate that there may be at least two classes of type II dlBNST neurons"

". In this study, we revealed the heterogeneity of type II dlBNST neurons."

(Asok et al., 2017)

Optogenetic silencing of a corticotropin-releasing factor pathway from the central amygdala to the bed nucleus of the stria terminalis disrupts sustained fear

"The lateral central nucleus of the amygdala (CeAL) and the dorsolateral bed nucleus of the stria terminalis (BNSTDL) coordinate the expression of shorter- and longer-lasting fears, respectively. Less is known about how these structures communicate with each other during fear acquisition. One pathway, from the CeAL to the BNSTDL, is thought to communicate via corticotropin-releasing factor (CRF), but studies have yet to examine its function in fear learning and memory. Thus, we developed an adeno-associated viral-based strategy to selectively target CRF neurons with the optogenetic silencer archaerhodopsin tp009 (CRF-ArchT) to examine the role of CeAL CRF neurons and projections to the BNSTDL during the acquisition of contextual fear. Expression of our CRF-ArchT vector injected into the amygdala was restricted to CeAL CRF neurons. Furthermore, CRF axonal projections from the CeAL clustered around BNSTDL CRF cells. Optogenetic silencing of CeAL CRF neurons during contextual fear acquisition disrupted retention test freezing 24 h later, but only at later time points (>6 min) during testing. Silencing CeAL CRF projections in the BNSTDL during contextual fear acquisition produced a similar effect. Baseline contextual freezing, the rate of fear acquisition, freezing in an alternate context after conditioning and responsivity to foot shock were unaffected by optogenetic silencing. Our results highlight how CeAL CRF neurons and projections to the BNSTDL consolidate longer-lasting components of a fear memory. Our findings have implications for understanding how discrete amygdalar CRF pathways modulate longer-lasting fear in anxiety- and trauma-related disorders."

(Arakaki et al., 2021)

Histological analyses of CRF mRNA expression in the dorsolateral bed nucleus of the stria terminalis of chronic pain model rats

"We previously reported that VTA-projecting dorsolateral bed nucleus of the stria terminalis (dlBNST) neurons make synapses onto the VTA GABAergic neurons"

"Recently, using electrophysiological analyses, we found the increased inhibitory inputs to VTA-projecting dlBNST neurons mediated by enhanced CRF signaling in the dlBNST of chronic pain model rats. CRF production has been reported to be produced in both intrinsic neurons within the dlBNST and the neurons projecting from the central amygdala to the dlBNST. Therefore, in this study, CRF mRNA expression in these brain areas of chronic pain model rats was examined by in situ hybridization histochemistry (ISH). CRF mRNA expression was increased in both dlBNST and CeA."

(Giardino et al., 2018)

PARALLEL CIRCUITS FROM THE BED NUCLEI OF STRIA TERMINALIS TO THE LATERAL HYPOTHALAMUS DRIVE OPPOSING EMOTIONAL STATES

"Lateral hypothalamus (LH) neurons containing the neuropeptide hypocretin (HCRT; orexin) modulate affective components of arousal, but their relevant synaptic inputs remain poorly defined. Here we identified inputs onto LH neurons that originate from neuronal populations in the bed nuclei of stria terminalis (BNST; a heterogeneous region of extended amygdala). We characterized two non-overlapping LH-projecting GABAergic BNST subpopulations that express distinct neuropeptides (corticotropin-releasing factor, CRF, and cholecystokinin, CCK). To functionally interrogate BNST LH circuitry, we used tools for monitoring and manipulating neural activity with cell-type-specific resolution in freely behaving mice. We found that Crf-BNST and Cck-BNST neurons respectively provide abundant and sparse inputs onto Hcrt-LH neurons, display discrete physiological responses to salient stimuli, drive opposite emotionally valenced behaviors, and receive different proportions of inputs from upstream networks. Together, our data provide an advanced model for how parallel BNST LH pathways promote divergent emotional states via connectivity patterns of genetically defined, circuit-specific neuronal subpopulations. Using genetically encoded physiological tools for monitoring, manipulating, and mapping discrete neural circuits, Giardino et al. characterize 2 discrete amygdala hypothalamus pathways that promote opposite behavioral responses to emotional stimuli."

(Romero et al., 2026)

Corticotropin-releasing factor neurons in the bed nucleus of the stria terminalis modulate avoidance behaviors and feeding

"Anatomical tracing revealed that BNST CRF neurons project to key hypothalamic regions, including the lateral hypothalamus (LH) and paraventricular hypothalamic nucleus (PVH)"

".The BNST projects broadly to brain regions involved in autonomic regulation, energy homeostasis, rewards, and motivation-including the hypothalamus, brainstem, and nucleus accumbens-positioning it at an anatomical crossroads linking anxiety,

motivation, and metabolic regulation (Sun et al., 2023)(Hao et al., 2019)(Giardino et al., 2018)(Kim et al., 2013). Prior anatomical tracing studies, including those catalogued in the Allen Brain Connectivity Atlas (Projection Experiment Page::Allen Brain Atlas: Mouse Connectivity and Projection Experiment Page::Allen Brain Atlas: Mouse Connectivity), have identified several projection targets of BNST CRF neurons, notably the paraventricular hypothalamic nucleus (PVH), lateral hypothalamic nucleus (LH) and amygdala (Oh et al., 2014)"

".histological analysis revealed robust synaptophysin GFP-positive terminal fields in both the LH (Figure 4B) and PVH (Figure 4C), consistent with previously reported connectivity."

(Sun et al., 2023)

Whole-brain input mapping of the lateral versus medial anterodorsal bed nucleus of the stria terminalis in the mouse

"The anterior portion of the bed nucleus of the stria terminalis (BNST) modulates fear and stress responses. The anterodorsal BNST (adBNST) can be anatomically subdivided further into the lateral and medial divisions. Although output projections of BNST subregions have been studied, the local and global input connections to these subregions remain poorly understood. To further understand BNST-centered circuit operations, we have applied new viral-genetic tracing and functional circuit mapping to determine detailed synaptic circuit inputs to lateral and medial subregions of adBNST in the mouse. Monosynaptic canine adenovirus type 2 (CAV2) and rabies virus-based retrograde tracers were injected in the adBNST subregions. The amygdalar complex, hypothalamus and hippocampal formation account for the majority of overall inputs to adBNST. However, lateral versus medial adBNST subregions have distinct patterns of long-range cortical and limbic brain inputs. The lateral adBNST has more input connections from prefrontal (prelimbic, infralimbic, cingulate) and insular cortices, anterior thalamus and ectorhinal/perirhinal cortices. In contrast, the medial adBNST received biased inputs from the medial amygdala, lateral septum, hypothalamus nuclei and ventral subiculum. We confirmed long-range functional inputs from the amydalohippocampal area and basolateral amygdala to the adBNST using ChR2-assisted circuit mapping. Selected novel BNST inputs are also validated with the AAV axonal tracing data from the Allen Institute Mouse Brain Connectivity Atlas. Together, these results provide a comprehensive map of the differential afferent inputs to lateral and medial adBNST subregions, and offer new insight into the functional operations of BNST circuitry for stress and anxiety-related behaviors."

(Kim et al., 2013)

Diverging neural pathways assemble a behavioural state from separable features in anxiety

"Behavioural states in mammals, such as the anxious state, are characterized by several features that are coordinately regulated by diverse nervous system outputs, ranging from behavioural choice patterns to changes in physiology (in anxiety, exemplified respectively by risk-avoidance and respiratory rate alterations). Here we investigate if and how defined neural projections arising from a single coordinating brain region in mice could mediate diverse features of anxiety. Integrating behavioural assays, in vivo and in vitro electrophysiology, respiratory physiology and optogenetics, we identify a surprising new role for the bed nucleus of the stria terminalis (BNST) in the coordinated modulation of diverse anxiety features. First, two BNST subregions were unexpectedly found to exert opposite effects on the anxious state: oval BNST activity promoted several independent anxious state features, whereas anterodorsal BNST-associated activity exerted anxiolytic influence for the same features. Notably, we found that three distinct anterodorsal BNST efferent projections—to the lateral hypothalamus, parabrachial nucleus and ventral tegmental area—each implemented an independent feature of anxiolysis: reduced risk-avoidance, reduced respiratory rate, and increased positive valence, respectively. Furthermore, selective inhibition of corresponding circuit elements in freely moving mice showed opposing behavioural effects compared with excitation, and in vivo recordings during free behaviour showed native spiking patterns in anterodorsal BNST neurons that differentiated safe and anxiogenic environments. These results demonstrate that distinct BNST subregions exert opposite effects in modulating anxiety, establish separable anxiolytic roles for different anterodorsal BNST projections, and illustrate circuit mechanisms underlying selection of features for the assembly of the anxious state."

Functions and behavioral relevance

dlBNST/BNST CRF neurons are best understood as stress- and aversion-linked modulators rather than as simple output cells with one fixed job. The clearest functional evidence here is that CRF signaling in dlBNST changes under chronic pain and can reshape the BNST VTApathway in a way that is well placed to alter negative affect and motivation. (1 source)

Functionally, dlBNST CRF neurons appear well suited to convert persistent aversive state into changes in downstream motivational circuitry. In chronic pain model rats, CRF signaling in the dlBNST was enhanced, and this was associated with **increased inhibitory input onto VTA-projecting dlBNST neurons**. Because these dlBNST projection neurons make synapses onto **VTA GABAergic neurons**, this circuit arrangement provides a concrete route by which dlBNST CRF activity can influence mesolimbic processing and thereby shape pain-related negative emotional states, motivational suppression, or aversion-linked behavior rather than only sensory aspects of pain (Arakaki et al., 2021).

The same study also showed that **CRF mRNA increased in both dlBNST and central amygdala** in chronic pain

animals, which supports a broader behavioral view of dlBNST CRF neurons as part of an extended amygdala stress system recruited during persistent adversity (Arakaki et al., 2021). Put simply, dlBNST CRF neurons are behaviorally relevant because they sit at a point where stress- and pain-related CRF signals can tune local inhibition and change how BNST outputs engage VTA-linked circuits, a role that fits well with broader BNST functions in negative affect, threat responding, and maladaptive state persistence (Arakaki et al., 2021) (Model-Generated).

Evidence

(Arakaki et al., 2021)

Histological analyses of CRF mRNA expression in the dorsolateral bed nucleus of the stria terminalis of chronic pain model rats

> *"We previously reported that VTA-projecting dorsolateral bed nucleus of the stria terminalis (dlBNST) neurons make synapses onto the VTA GABAergic neurons"*
>
> *"Recently, using electrophysiological analyses, we found the increased inhibitory inputs to VTA-projecting dlBNST neurons mediated by enhanced CRF signaling in the dlBNST of chronic pain model rats. CRF production has been reported to be produced in both intrinsic neurons within the dlBNST and the neurons projecting from the central amygdala to the dlBNST. Therefore, in this study, CRF mRNA expression in these brain areas of chronic pain model rats was examined by in situ hybridization histochemistry (ISH). CRF mRNA expression was increased in both dlBNST and CeA."*

Single-cell transcriptomics and molecular diversity

Single-cell work argues that CRF neurons should not be treated as one molecularly fixed class. For dlBNST CRF neurons specifically, direct single-cell data are still limited in the provided papers, but BNST-wide transcriptomics and work in other CRF systems both point to substantial subtype structure, mixed transmitter identities, and likely state-dependent molecular variation. (7 sources)

Single-cell transcriptomics changes the default way to think about dlBNST CRF neurons. Rather than assuming that all CRF-expressing neurons in a region share one molecular program, single-cell studies in other CRF systems show that **Crh-positive neurons split into multiple transcriptional clusters** that can reflect both stable subpopulations and changing functional states. In developing mouse PVN, for example, single-cell analysis separated **CRF-expressing neurons into five clusters**, with some clusters mainly **glutamatergic** and others **GABAergic**, and the transcriptional response to early-life adversity was **highly subtype specific** rather than uniform across all CRF cells (Short et al., 2021) (Kim et al., 2019) (Romanov et al., 2020) (Xu et al., 2020). This does not describe BNST directly, but it provides an important interpretive frame: a peptide label like **Crh/CRF** is likely only the first layer of identity, not the full cell type definition.

For the BNST itself, the key transcriptomic message is that it is **exceptionally diverse at single-cell resolution**. Mouse BNST single-cell RNA-seq identified **41 transcriptionally distinct cell types**, with about **86% GABAergic** and **14% glutamatergic** cells, and later summaries emphasize high transcriptional diversity in both rodent and human BNST (Ly et al., 2025) (Welch et al., 2019) (Siletti et al., 2022). That broad BNST map fits well with the earlier physiological and anatomical evidence that dlBNST CRF neurons are embedded in a dense local mosaic. It also suggests that the dlBNST CRF population is unlikely to be a single transcriptomic cluster; more plausibly, it spans several related GABAergic states or subtypes, with smaller glutamatergic CRF subsets depending on exact subregion.

A practical implication is that future definitions of dlBNST CRF neurons will likely need **combinatorial markers**, not CRF alone. Across hypothalamic single-cell studies, molecular identity is best resolved by combinations of **neuropeptides, transmitter genes, receptors, and transcription factors**, and functional work in PVH shows that quantitative gene-expression patterns can predict neuronal response profiles better than a one-marker label (Romanov et al., 2020) (Xu et al., 2020). For BNST, this means the most useful dlBNST CRF taxonomy will probably combine **Crh** with transmitter markers and region- or state-linked genes, then tie those profiles back to projection targets and electrophysiology. In short, the available single-cell evidence supports a view of dlBNST CRF neurons as a **molecularly diverse family within a highly heterogeneous BNST**,

not a single uniform neuropeptidergic cell type (Ly et al., 2025) (Welch et al., 2019).

Evidence

(Short et al., 2021)

Single-Cell Transcriptional Changes in Hypothalamic Corticotropin-Releasing Factor–Expressing Neurons After Early-Life Adversity Inform Enduring Alterations in Vulnerabilities to Stress

"In contrast to this traditional classification, recent single-cell analyses of hypothalamic transcriptomes have identified multiple molecular-defined clusters of CRFexpressing neurons that may represent different subpopulations or distinct neuronal functional states (Kim et al., 2019)(Romanov et al., 2017)(Romanov et al., 2020)(Xu et al., 2020)"

".We used single-cell RNA sequencing to probe the effects of ELA on gene expression programs in distinct neuronal populations, determine the potential selectivity of the effects of ELA, and identify their downstream consequences"

".The hypothalamic PVN harbors several types of CRFexpressing neurons, including neuroendocrine cells releasing CRF into the portal bloodstream and others projecting to specific brain areas (Swanson et al., 1987)(Aguilera et al., 2011). The majority of PVN CRFexpressing (CRF1) cells coexpress glutamate, a key excitatory neurotransmitter (Hrabovszky et al., 2008)(Hrabovszky et al., 2005), whereas others express genes associated with inhibitory GABAergic (gamma-aminobutyric acidergic) neurotransmission (Meister et al., 1988). To delineate these populations in developing mouse PVN and determine their potential contribution to the phenotypic consequences of ELA, we used single-cell transcriptomics"

".Focusing on cells coexpressing Crh and the neuronal synaptic protein encoding gene Snap25 (Figure 1D), we characterized them using shared nearest neighbor clustering (using the Seurat package) (62), which segregated CRF1 neurons into five distinct clusters (Figure 1E)"

".The expression-based clustering identified patterns of gene expression that suggested that clusters 1, 3, and 4 are primarily glutamatergic and clusters 2 and 5 are GABAergic (for detailed summary, see Figure S3B)"

".Differential expression analysis between neurons from the ELA and control groups within each neurotransmitter-defined cluster revealed that differentially expressed genes (false discovery rate , 0.1) between control and ELA cells were highly subtype specific (Figure 3B). There were no differentially expressed genes in the GABAergic cluster or that comprised cells expressing no neurotransmitter markers."

(Kim et al., 2019)

The cellular and molecular landscape of hypothalamic patterning and differentiation from embryonic to late postnatal development

"The hypothalamus is a central regulator of many innate behaviors essential for survival, but the molecular mechanisms controlling hypothalamic patterning and cell fate specification are poorly understood. To identify genes that control hypothalamic development, we have used single-cell RNA sequencing (scRNA-Seq) to profile mouse hypothalamic gene expression across 12 developmental time points between embryonic day 10 and postnatal day 45. This identified genes that delineated clear developmental trajectories for all major hypothalamic cell types, and readily distinguished major regional subdivisions of the developing hypothalamus. By using our developmental dataset, we were able to rapidly annotate previously unidentified clusters from existing scRNA-Seq datasets collected during development and to identify the developmental origins of major neuronal populations of the ventromedial hypothalamus. We further show that our approach can rapidly and comprehensively characterize mutants that have altered hypothalamic patterning, identifying Nkx2.1 as a negative regulator of prethalamic identity. These data serve as a resource for further studies of hypothalamic development, physiology, and dysfunction. The cellular and molecular mechanisms regulating hypothalamic patterning and differentiation are unclear. Here, the authors profiled the transcriptome of the developing hypothalamus at single cell level, providing a resource to hypothalamic development in health and disease."

(Romanov et al., 2020)

Molecular design of hypothalamus development

"A wealth of specialized neuroendocrine command systems intercalated within the hypothalamus control the most fundamental physiological needs in vertebrates1,2. Nevertheless, we lack a developmental blueprint that integrates the molecular determinants of neuronal and glial diversity along temporal and spatial scales of hypothalamus development3. Here we combine single-cell RNA sequencing of 51,199 mouse cells of ectodermal origin, gene regulatory network (GRN) screens in conjunction with genome-wide association study-based disease phenotyping, and genetic lineage reconstruction to show that nine glial and thirty-three neuronal subtypes are generated by mid-gestation under the control of distinct GRNs. Combinatorial molecular codes that arise from neurotransmitters, neuropeptides and transcription factors are minimally required to decode the taxonomical hierarchy of hypothalamic neurons. The differentiation of γ-aminobutyric acid (GABA) and dopamine neurons, but not glutamate neurons, relies on quasi-stable intermediate states, with a pool of GABA progenitors giving rise to dopamine cells4. We found an unexpected abundance of chemotropic proliferation and guidance cues that are commonly implicated in dorsal (cortical) patterning5 in the hypothalamus. In particular, loss of SLIT–ROBO signalling impaired both the production and positioning of periventricular dopamine

neurons. Overall, we identify molecular principles that shape the developmental architecture of the hypothalamus and show how neuronal heterogeneity is transformed into a multimodal neural unit to provide virtually infinite adaptive potential throughout life. Single-cell RNA sequencing reveals molecular determinants of the developmental programs that orchestrate the intermingling of neuronal subtypes in the hypothalamus."

(Xu et al., 2020)

Behavioral state coding by molecularly defined paraventricular hypothalamic cell type ensembles

"How neuron types encode behavioral states What is the contribution of molecularly defined cell types to neural coding of stimuli and states? Xu et al. aimed to evaluate neural representation of multiple behavioral states in the mouse paraventricular hypothalamus. To achieve this goal, they combined deep-brain two-photon imaging with post hoc validation of gene expression in the imaged cells. The behavioral states could be well predicted by the neural response of multiple neuronal clusters. Some clusters were broadly tuned and contributed strongly to the decoding of multiple behavioral states, whereas others were more specifically tuned to certain behaviors or specific time windows of a behavioral state. Science, this issue p. eabb2494 An imaging method can merge molecular and systems neuroscience to reveal combinatorial cell type coding of essential survival behaviors. INTRODUCTION Brain function is often compared to an orchestral ensemble, where subgroups of neurons that have similar activity are analogous to different types of instruments playing a musical score. Brains are composed of specialized neuronal subtypes that can be efficiently classified by gene expression profiles measured by single-cell RNA sequencing (scRNA-seq). Are these molecularly defined cell types the "instruments" in the neural ensemble? To address this question, we examined the neural ensemble dynamics of the hypothalamic paraventricular nucleus (PVH), a small brain region that is important for behavior states such as hunger, thirst, and stress. Past work has emphasized specialized behavioral state–setting roles for different PVH cell types, but it is not clear whether the dynamics of the PVH ensemble support this view. RATIONALE We considered three possibilities for how PVH neurons could be involved in encoding behavioral states: (i) PVH neurons of a molecularly defined cell type may respond similarly and be specialized for a behavioral state as a "labeled-line," (ii) molecularly defined cell types may show unrelated activity patterns and be irrelevant to behavioral state coding, and (iii) molecularly defined neurons may respond similarly within a type, but behavioral state may be encoded by combinations of cell types. To evaluate the role of molecularly defined cell types in the neural ensemble, it is important to monitor activity in many individual neurons with subsecond temporal resolution along with quantitative gene expression information about each cell. For this, we developed the CaRMA (calcium and RNA multiplexed activity) imaging platform in which deep-brain two-photon calcium imaging of neuron activity is performed in mice during multiple behavioral tasks. This is followed by ex vivo multiplexed RNA fluorescent in situ hybridization to measure gene expression information in the in vivo–imaged neurons. RESULTS We simultaneously imaged calcium activity in hundreds of PVH neurons from 10 cell types across 11 behavioral states. Within a molecularly defined cell type, neurons often showed similar activity patterns such that we could predict functional responses of individual neurons solely from their quantitative gene expression information. Behavioral states could be decoded with high accuracy based on combinatorial assemblies of PVH cell types, which we called "grouped-ensemble coding." Labeled-line coding was not observed. The neuromodulatory receptor gene neuropeptide receptor neuropeptide Y receptor type 1 (Npy1r) was usually the most predictive gene for neuron functional response and was expressed in multiple cell types, analogous to the "conductor" of the PVH neural ensemble. CONCLUSION Our results validated molecularly defined neurons as important information processing units in the PVH. We found correspondence between the gene expression hierarchies used for molecularly defined cell type classification and functional activity hierarchies involving coordination by neuromodulation. CaRMA imaging offers a solution to the problem of how to rapidly evaluate the function of the panoply of cell types being uncovered with scRNA-seq. CaRMA imaging bridges a gap between the abstract digital elements typically described in systems neuroscience with the "wetware" associated with traditional molecular neuroscience. Merging these two areas is essential to understanding the relationships of gene expression, brain function, behavior, and ultimately neurological diseases. CaRMA imaging reveals combinatorial cell type coding of behavior states. CaRMA imaging records calcium dynamics of PVH neurons across multiple behavioral states followed by gene expression profiling. Combinatorial assemblies of PVH cell types encoded behavioral states. The PVH neural activity ensemble was split by Npy1r expression into two main cell classes that were subdivided into cell types. Thus, neuromodulation coordinates cell types for grouped-ensemble coding to represent different survival behaviors such as eating, drinking, and stress. Brains encode behaviors using neurons amenable to systematic classification by gene expression. The contribution of molecular identity to neural coding is not understood because of the challenges involved with measuring neural dynamics and molecular information from the same cells. We developed CaRMA (calcium and RNA multiplexed activity) imaging based on recording in vivo single-neuron calcium dynamics followed by gene expression analysis. We simultaneously monitored activity in hundreds of neurons in mouse paraventricular hypothalamus (PVH). Combinations of cell-type marker genes had predictive power for neuronal responses across 11 behavioral states. The PVH uses combinatorial assemblies of molecularly defined neuron populations for grouped-ensemble coding of survival behaviors. The neuropeptide receptor neuropeptide Y receptor type 1 (Npy1r) amalgamated multiple cell types with similar responses. Our results show that molecularly defined neurons are important processing units for brain function."

(Ly et al., 2025)

VGluT3 BNST neurons transmit GABA and restrict sucrose consumption

"With novel tools to probe the transcriptomics of neuroanatomically distinct brain regions, it was discovered that the human and rodent BNST have a high level of transcriptional diversity (Siletti et al., 2022)(Welch et al., 2019). Single-cell RNA sequencing data from mouse BNST revealed 41 transcriptionally distinct cell-types with 86% of cells expressing GABAergic genetic markers and the remaining 14% expressing glutamatergic genetic markers (Welch et al., 2019)"

".We identified a unique cell-type within the anteromedial BNST that expressed the vesicular glutamate transporter type 3 (VGluT3), a molecular marker for the vesicular loading of glutamate into synaptic vesicles. BNST VGluT3 neurons were distinct from striatal VGluT3 neurons due to the absence of the cholinergic marker choline acetyltransferase that classify striatal VGluT3-expressing cholinergic interneurons [45e[47]. VGluT3 BNST neurons mostly resided in the anteromedial BNST region and most single BNST VGluT3 neurons projected to either PVN or ARC with a subset that projected to both hypothalamic regions."

(Welch et al., 2019)

Single-cell multi-omic integration compares and contrasts features of brain cell identity

"Summary Defining cell types requires integrating diverse single-cell measurements from multiple experiments and biological contexts. To flexibly model single-cell datasets, we developed LIGER, an algorithm that delineates shared and dataset-specific features of cell identity. We applied it to four diverse and challenging analyses of human and mouse brain cells. First, we defined region-specific and sexually dimorphic gene expression in the mouse bed nucleus of the stria terminalis. Second, we analyzed expression in the human substantia nigra, comparing cell states in specific donors and relating cell types to those in the mouse. Third, we integrated in situ and single-cell expression data to spatially locate fine subtypes of cells present in the mouse frontal cortex. Finally, we jointly defined mouse cortical cell types using single-cell RNA-seq and DNA methylation profiles, revealing putative mechanisms of cell-type-specific epigenomic regulation. Integrative analyses using LIGER promise to accelerate investigations of cell-type definition, gene regulation, and disease states."

(Siletti et al., 2022)

Transcriptomic diversity of cell types across the adult human brain

"The human brain directs a wide range of complex behaviors ranging from fine motor skills to abstract intelligence and emotion. However, the diversity of cell types that support these skills has not been fully described. Here we used high-throughput single-nucleus RNA sequencing to systematically survey cells across the entire adult human brain in three postmortem donors. We sampled over three million nuclei from approximately 100 dissections across the forebrain, midbrain, and hindbrain. Our analysis identified 461 clusters and 3313 subclusters organized largely according to developmental origins. We found area-specific cortical neurons, as well as an unexpectedly high diversity of midbrain and hindbrain neurons. Astrocytes also exhibited regional diversity at multiple scales, comprising subtypes specific to the telencephalon and to more precise anatomical locations. Oligodendrocyte precursors comprised two distinct major types specific to the telencephalon and to the rest of the brain. Together, these findings demonstrate the unique cellular composition of the telencephalon with respect to all major brain cell types. As the first single-cell transcriptomic census of the entire human brain, we provide a resource for understanding the molecular diversity of the human brain in health and disease."